



Original contribution

The effect of ultrasound-guided intercostal nerve block, single-injection erector spinae plane block and multiple-injection paravertebral block on postoperative analgesia in thoracoscopic surgery: A randomized, double-blinded, clinical trial[☆]

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ABSTRACT

Study objective: The study was to determine the analgesic effect of ultrasound-guided intercostal nerve block (ICNB) and single-injection erector spinae plane block (ESPB) in comparison with multiple-injection paravertebral block (PVB) after thoracoscopic surgery.

Design: Randomized, controlled, double-blinded study.

Setting: Operating room, postoperative recovery room and ward.

Patients: Seventy-five patients, aged 18–75 years, ASA I–II and scheduled for elective thoracoscopic partial pulmonary resection surgery were enrolled in the study. Seventy-two patients were left for final analysis.

Interventions: Patients were randomly assigned into the three groups (PVB group, ICNB group or ESPB group). After anesthesia induction, a single anesthesiologist performed PVB at T5–T7 levels or ICNB at T4–T9 levels or ESPB at T5 level under ultrasound guidance using 20 ml of 0.375% ropivacaine. Patients were connected to the patient-controlled morphine analgesia device after surgery.

Measurements: Cumulative morphine consumption at 24 h postoperatively as primary outcome was compared. Visual analog scale pain scores at rest and while coughing at 0, 2, 4, 8, 24 and 48 h postoperatively, cumulative morphine consumption at other observed time and rescue analgesia requirement were also recorded.

Main results: There was a significant difference in median [interquartile range, IQR] morphine consumption at 24 h postoperatively among the three groups (PVB, 10.5 [9–15] mg; ICNB, 18 [13.5–22.1] mg; ESPB, 22 [15–25.1] mg; $p = 0.000$). This difference was statistically significant for PVB group vs ESPB group (median difference, -7.5 ; 95% confidence interval [CI], -12 to -4.5 ; $p = 0.000$) and PVB group vs ICNB group (median difference, -6 ; 95% CI, -9 to -3 ; $p = 0.001$), but not for ICNB vs ESPB (median difference, -3 ; 95% CI, -6 to 1.5 ; $p = 0.192$). PVB group had significantly lower VAS scores at rest and while coughing than ESPB group at 0, 2, 4, 8 h postoperatively and than ICNB group at 8 h postoperatively. There was no significant difference in the VAS scores between ICNB group and ESPB group at all time. Median VAS scores at rest and while coughing at all time were low (< 4) in all groups. More rescue analgesia was needed in ESPB group during 48 postoperative hours (PVB vs ICNB vs ESPB; 13% vs 29% vs 46%; $p < 0.05$).

Conclusions: Ultrasound-guided multiple-injection PVB provided superior analgesia to ICNB and single-injection ESPB, while ICNB and single-injection ESPB were equally effective in reducing pain after thoracoscopic surgery.

1. Introduction

Thoracoscopic surgery has been used widely because it has several advantages over conventional thoracotomy, such as less postoperative complications, less postoperative pain and shorter hospital stay [1,2].

Even though it is considered as a less invasive procedure, patients after thoracoscopic surgery also experience moderate-to-severe pain. Inadequate analgesia will increase patients' suffering and make patients unable to breathe normally and have an ineffective cough, which will increase respiratory complications and affect postoperative recovery.

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Recent years, regional anesthesia techniques play a more important role in multimodal analgesia due to reducing postoperative pain and opioids usage and related side effects. However, the optimal regional anesthesia technique, which is expected to be effective, safe, simple and less invasive, remains a debate after thoracoscopic surgery.

Though epidural analgesia was once considered as the gold standard for post-thoracotomy pain management, it is not recommended for pain control after thoracoscopic surgery because it is associated with high potential risks of dural puncture, nerve lesions, epidural hematoma and hypotension. Thoracic paravertebral block (PVB) can provide comparable analgesia to epidural analgesia with more favorable side-effects [3–5], which is another gold standard after thoracotomy. However, it is not in regular use because of technical challenge and potential risks. Intercostal nerve block (ICNB) has been reported to be effective in pain control after thoracotomy [6–8]. It is utilized mainly due to its technical safety and simplicity. But it necessitates multiple injections. More recently, a novel interfascial plane block, ultrasound-guided erector spinae plane block (ESPB) has been described as an effective post-operative analgesia technique in various surgeries [9–12]. As reported, local anesthetics (LA) in ESPB expected to extend to paravertebral space and acted on both the ventral and dorsal rami of the spinal nerves [13]. Several studies have shown ESPB performed at the level of the T5 transverse process was capable of providing extensive thoracic analgesia, from T3 to T9 over the hemithorax [13]. In addition, transverse process as an endpoint for needle advancement, which is a identifiable ultrasonographic landmark and far away from the pleura and neuraxial structures, makes ESPB easy to perform and minimizes risks of complication. Hence, it was speculated that ESPB is promising to be a simple alternative to PVB [14].

Studies regarding the analgesic effect of ICNB after thoracoscopic surgery in comparison with paravertebral block are scarce. Also analgesic effect of ESPB after thoracoscopic surgery has not yet been clear and there is no direct evidence to support the hypothesis that ESPB can be an alternative to PVB because there were no randomized controlled trials comparing ESPB with PVB or other analgesic techniques in thoracoscopic surgery. Therefore, we designed the study to determine the postoperative analgesic efficacy of ultrasound-guided ICNB and single-injection ESPB in comparison with multiple-injection paravertebral block in patients undergoing thoracoscopic surgery and confirm whether ESPB can be an alternative to PVB.

2. Methods

2.1. Study design and randomization

This prospective, randomized controlled study was approved by the Ethics Committee of the Tongji Hospital affiliated to Tongji Medical College of Huazhong University of Science and Technology (TJ-IRB20180903) and was registered to Chinese Clinical Trials Registry (Registration No: ChiCTR1800018672). Recruitment was performed between October 2018 and February 2019. Written informed consents were obtained from all patients. The CONSORT flow diagram was used for enrollment and allocation of patients (Fig. 1).

Patients aged 18–75 years with American Society of Anesthesiologists (ASA) physical status I–II and undergoing elective thoracoscopic partial pulmonary resection surgery in our institute were enrolled in the study. The exclusion criteria included allergy to local anesthetics, coagulation disorders, severe heart diseases, hepatic or renal insufficiency, a history of chronic pain or chronic opioid use and psychiatric disease.

Patients were allocated randomly to PVB group, ICNB group and ESPB group according to computer-generated random number table. Allocation to treatment group was performed using the sealed opaque envelope technique. All patients and an investigator who was responsible for follow-up during 48 postoperative hours were blinded to the randomization groups. In addition, during preoperative visits, we

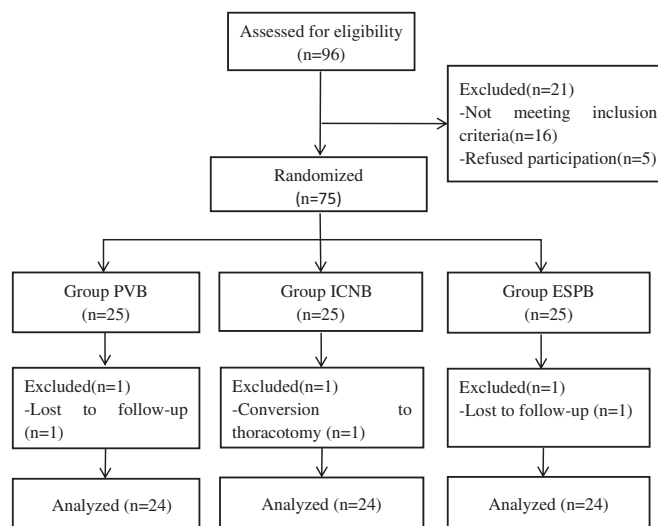


Fig. 1. Consort flow diagram of patients.

instructed patients how to use the patient-controlled intravenous analgesia (PCA) device for pain management, as well as how to use the visual analog scale to evaluate pain at rest and while coughing.

2.2. General anesthesia and surgical technique

Patients were monitored using an electrocardiogram, pulse oximetry and non-invasive blood pressure (one measurement every 3 min) while entering operation room. A radial artery catheter was also placed for invasive arterial pressure and blood gas monitoring. Parecoxib sodium 40 mg, penehyclidine hydrochloride 0.5 mg and tropisetron 2 mg were given intravenously before anesthesia for preemptive analgesia, reduction of secretion and antiemetic prophylaxis. General anesthetic induction was conducted with sufentanil $0.5 \mu\text{g kg}^{-1}$, propofol $1.5\text{--}2.0 \text{ mg kg}^{-1}$ and rocuronium 0.8 mg kg^{-1} . Endotracheal intubation was performed with a double-lumen tube. Anesthesia was maintained with inhaled sevoflurane 1.5% in 40% oxygen in air and intravenous infusion of dexmedetomidine at $0.5 \mu\text{g kg}^{-1} \text{ h}^{-1}$ and remifentanyl at $0.1\text{--}0.2 \mu\text{g kg}^{-1} \text{ min}^{-1}$. After general anesthesia, placing the patient in the proper position and completing standard skin disinfection and wrapping probe with sterile membrane, a single anesthesiologist with significant ultrasound-guided regional anesthesia experience performed all blocks. Thoracoscopic surgery was performed via a two-port technique. A 3-cm operation port and a 1.5-cm observation port were respectively made in the fifth intercostal space at the anterior axillary line and in the seventh or eighth intercostal space at posterior axillary line. Approximately 30 min before the end of surgery, dezocine 5 mg was given intravenously. Two chest tubes were inserted into the two incisions for air leakage and drainage after surgery. Patients were extubated at the end of surgery when were fully awake and breathed adequately. Then, they were transferred to the postanesthesia care unit (PACU) for a 1-h observation period.

2.3. Ultrasound-guided paravertebral block

The patient was placed in the lateral decubitus position. A linear L15-4B ultrasound probe (Wisonic Navi s, Shenzhen, China) was oriented longitudinally on the patient's midline at the T5 level. The probe was then moved laterally (3 cm from the midline) until an appropriate view of adjacent transverse processes, corresponding paravertebral space and pleura. An 18-gauge, 10-cm needle (Stimuplex A, Braun, Melsungen, Germany) was inserted via an in-plane technique and pierced superior costotransverse ligament into the paravertebral space. After negative aspiration, 6.7 ml of 0.375% ropivacaine was injected.

This procedure was repeated at T6 and T7 level with 6.6 ml and 6.7 ml of 0.375% ropivacaine respectively. When pleural displacement was observed at all three levels, PVB was seen to be successful.

2.4. Ultrasound-guided intercostal nerve block

The patient was in supine position. Unilateral ICNB was performed at the levels of T4–T9. A linear L15-4B ultrasound probe (Wisonic Navi s, Shenzhen, China) was placed in a longitudinal orientation to identify the rib, internal intercostal muscles, innermost intercostal muscles and pleura. An 18-gauge, 10-cm needle (Stimuplex A, Braun, Melsungen, Germany) was inserted to target the inferior margin of the rib in an in-plane approach. After negative aspiration, the anesthesiologist injected 4 ml of 0.375% ropivacaine into the intercostal spaces where two incisions located and 3 ml ropivacaine for other levels. When pleural displacement was observed in all intercostal spaces, ICNB was considered successful.

2.5. Ultrasound-guided erector spinae plane block

The patient was placed in the lateral decubitus position. Single-injection ESPB was performed at the T5 level of the spine. A linear L15-4B ultrasound probe (Wisonic Navi s, Shenzhen, China) was placed approximately 3 cm lateral to the midline and a paramedian sagittal scan was performed to identify the erector spinae muscle and the T5 transverse process. Then the anesthesiologist turned on the midline button of ultrasound apparatus and made the T5 transverse process in the middle of ultrasonic image. An 18-gauge, 10-cm needle (Stimuplex A, Braun, Melsungen, Germany) was inserted at the midpoint of the probe and advanced by an outplane approach until contacted the transverse process. After confirming the correct location of the needle tip by injecting 0.5–1 ml saline, 20 ml of 0.375% ropivacaine was injected into the interfascial plane between the erector spinae muscle and T5 transverse process. When local anesthetics spread in a linear pattern beneath the erector spinae muscle, we thought ESP block was successful.

2.6. Postoperative analgesia protocol and rescue analgesic

Upon arrival at PACU (0 h after surgery), patients were requested to evaluate pain at rest and while coughing using visual analog scale (VAS: 0 cm = no pain, 10 cm = worst pain imaginable). If VAS score was > 3 at rest (inadequate analgesia), oxycodone was administered intravenously as rescue analgesia before commencing the patient-controlled analgesia (PCA). Then all patients were connected to the PCA device (BCDB-200; BCM, Shanghai, China). The PCA device consisted of 0.5 mg ml⁻¹ morphine and was programmed as follow: 0.5 ml h⁻¹ background rate, 3 ml bolus doses and 5 min-lockout interval. In the ward, patients used PCA when VAS score at rest > 3 or on their demand. If three boluses of morphine did not alleviate pain, diclofenac sodium suppositories 50 mg were given as rescue analgesia.

2.7. Outcome measurements

The primary outcome was cumulative morphine consumption (rescue oxycodone was converted to intravenous morphine equivalents for final analysis) at 24 h postoperatively. The secondary outcomes included VAS scores at rest and while coughing at 0, 2, 4, 8, 24 and 48 h postoperatively, morphine consumption at other observed time, rescue analgesia requirement, block-related complications and postoperative adverse effects, such as pruritus, urinary retention, nausea and vomiting.

2.8. Sample size

The sample size was calculated on the basis of a pilot study that showed the 24 h postoperative morphine consumption was

12.0 ± 5.6 mg in the PVB group, 16.6 ± 6.55 mg in the ICNB group and 18.4 ± 5.0 mg in the ESPB group. Choosing a difference of 5 mg in 24 h postoperative morphine consumption as the minimum desired difference between the groups, 23 patients per group were required to achieve a significance level of 0.05 with a power of 80%. We planned to recruit 25 patients in each group to compensate for dropouts.

2.9. Statistical analysis

Statistical analyses were performed using SPSS version 22.0 software (IBM Corporation, USA). Continuous variables were presented as mean (SD) or median (interquartile range, [IQR]), depending on whether the data were distributed normally or not. Categorical variables were presented as number (percentage). Data distribution was evaluated using the Kolmogorov-Smirnov test. Demographic and perioperative data were analyzed using one-way analysis of variance (ANOVA) and χ^2 test. Nonnormal continuous outcomes, morphine consumption and VAS, were analyzed with Kruskal-Wallis test across groups. Moreover, pairwise comparisons were analyzed using the Mann-Whitney *U* test and *p* < 0.0167 (Bonferroni correction) was considered significant. The frequency of rescue analgesia, the incidence of block-related complications and postoperative adverse effects were compared with χ^2 test or Fisher's exact test. *p* < 0.05 was considered significant.

3. Results

Seventy-five patients were assessed for eligibility. Three patients were excluded from final analysis because one was converted to thoracotomy and two lost to follow-up. Therefore, seventy-two patients (24 patients in each group) completed the study (Fig. 1). The baseline demographics and perioperative variables were similar in the three groups (Table 1). There was no significant difference in the intraoperative remifentanyl and dexmedetomidine consumption.

The median [IQR] morphine consumption, including converted oxycodone, at 24 h postoperatively was significantly different among the three groups (PVB, 10.5 [9–15] mg; ICNB, 18 [13.5–22.1] mg; ESPB, 22 [15–25.1] mg; *p* = 0.000). This difference was statistically significant for PVB group vs ESPB group (median difference, -7.5; 95% confidence interval [CI], -12 to -4.5; *p* = 0.000) and PVB group vs ICNB group (median difference, -6; 95% CI, -9 to -3; *p* = 0.001), but not for ICNB vs ESPB (median difference, -3; 95% CI, -6 to 1.5;

Table 1
Demographic and perioperative characteristics.

	PVB (n = 24)	ICNB (n = 24)	ESPB (n = 24)
Sex			
F	11 (46%)	9 (38%)	11 (46%)
M	13 (54%)	15 (62%)	13 (54%)
Age (yr)	51.6 (10.4)	58.1 (7.0)	53.3 (11.6)
Weight (kg)	63.6 (6.3)	64.6 (9.5)	62.0 (10.9)
Height (cm)	166.8 (7.6)	165.5 (8.1)	164.2 (7.8)
BMI (kg/m ²)	22.9 (2.6)	23.5 (2.4)	22.9 (3.1)
ASA class			
I	9 (38%)	9 (38%)	10 (42%)
II	15 (62%)	15 (62%)	14 (58%)
Type of VATS			
Lobectomy	15 (63%)	18 (75%)	17 (71%)
Segmentectomy	1 (4%)	1 (4%)	1 (4%)
Wedge resection	8 (33%)	5 (21%)	6 (25%)
Duration of surgery (min)	128.4 (58.2)	144.8 (46.0)	134.5 (43.1)
Total remifentanyl (μg)	892.1 (269.3)	934.2 (229.1)	923.0 (354.3)
Total dexmedetomidine (μg)	46.6 (20.7)	51.6 (17.4)	49 (21.1)

Data are presented as mean (SD) or number (%). PVB = paravertebral block; ICNB = intercostal nerve block; ESPB = erector spinae plane block; BMI = body mass index; F = female; M = male; ASA = American Society of Anesthesiologists.

Table 2
Morphine consumption.

	PVB	ICNB	ESPB	Pairwise comparisons		
	(n = 24)	(n = 24)	(n = 24)	PVB vs. ICNB Median difference (95% CI) p value	PVB vs. ESPB Median difference (95% CI) p value	ICNB vs. ESPB Median difference (95% CI) p value
2 h	0.5 (0.5–0.5)	0.5 (0.5–2)	0.5 (0.5–3.1)	0 (0 to 0) 0.077	0 (–1.5 to 0) 0.014*	0 (–0.5 to 0) 0.324
4 h	1 (1–2.5)	2.5 (1–5.5)	2.5 (1–7.8)	–0.5 (–1.5 to 0) 0.036	–1.5 (–3 to 0) 0.014*	0 (–1.5 to 1) 0.566
8 h	3.5 (2–5)	5.75 (5–11)	8 (5–12.5)	–3 (–4.5 to –1.5) 0.001*	–3 (–7.5 to –1.5) 0.000*	–1.5 (–3 to 1.5) 0.375
24 h	10.5 (9–15)	18 (13.5–22.1)	22 (15–25.1)	–6 (–9 to –3) 0.001*	–7.5 (–12 to –4.5) 0.000*	–3 (–6 to 1.5) 0.192
48 h	18 (15–21)	27.8 (22.5–30)	29.5 (22.5–32.6)	–7.5 (–10.5 to –4.5) 0.000*	–9 (–12.5 to –4.5) 0.000*	–1.5 (–5 to 1.5) 0.336

Data are presented as median (IQR). All groups were compared using Kruskal-Wallis test. Pairwise comparisons were analyzed using Mann-Whitney *U* test and **p* < 0.0167 (Bonferroni correction). PVB = paravertebral block; ICNB = intercostal nerve block; ESPB = erector spinae plane block; CI: interquartile range.

p = 0.192). Moreover, PVB group had significantly lower morphine consumption compared with ESPB group at all observed time and with ICNB group at 8, 24, 48 h postoperatively. No significant difference in morphine consumption was found between ICNB group and ESPB group at all time points (Table 2).

As indicated in Figs. 2 and 3, VAS scores at rest and while coughing were statistically significantly lower for PVB group compared to ESPB group at 0, 2, 4 and 8 h postoperatively and to the ICNB group at 8 h postoperatively. However, median VAS scores at rest and while coughing were low (< 4) at all time in all groups. There was also no significant difference in VAS scores between ESPB group and ICNB group.

The rescue analgesia requirement in these three groups was shown in Table 3. The frequency of rescue analgesia during 48 postoperative hours was different across groups (PVB vs ICNB vs ESPB; 13% vs 29% vs 46%; *p* = 0.040). More rescue analgesia was needed in ESPB group compared to PVB group (ESPB vs PVB, *p* = 0.011; PVB vs ICNB, *p* = 0.155; ESPB vs ICNB, *p* = 0.233). Hematoma was noted in four patients in PVB group, five in ICNB group and none in ESPB group after performance of block (*p* = 0.660). Besides, four, three and five patients respectively in PVB group, ICNB group and ESPB group had nausea or vomiting (*p* = 0.922). No patients developed pneumothorax, pruritus and urinary retention in all groups.

4. Discussion

Our study showed ultrasound-guided multiple-injection PVB provided superior analgesia for thoracoscopic surgery to ICNB and single-

injection ESPB, while ICNB and single-injection ESPB were equally effective in reducing pain. Notably, to our knowledge, this was the first randomized controlled study to compare ESPB with PVB and ICNB on analgesic effect after thoracoscopic surgery.

Our present study indicated patients in ICNB group and ESPB group had significantly more morphine consumption than those in PVB group at 24 h postoperatively. However, in clinical practice, the difference in morphine consumption seemed relatively small. Three reasons may account for the results. First, it was probably that good manipulation abilities and skills of surgeons caused less pain than expected. In more invasive surgery, larger difference in morphine consumption between ICNB and PVB or ESPB and PVB would reveal so that significant analgesic superiority of PVB would be showed. Second, it is difficult for patients to effectively operate PCA device. Furthermore, in China, patients are unaware of the importance of effective pain management after surgery and think pain is the inevitable result of surgery. So patients are more stoic and seldom press the PCA button despite experiencing severe pain [15], which may lead morphine consumption in ICNB and ESPB group was lower than that found by other studies. Thirdly, more rescue analgesic used in ICNB group and ESPB group decreased morphine consumption. In these situations, that respectively 52% and 42% reduction in morphine consumption at 24 h postoperatively in PVB group compared to the ESPB group and ICNB group showed our trial had certain clinical significance.

It was reported analgesic duration of PVB after thoracoscopic surgery could last 6–48 h [16,17]. VAS scores and morphine consumption did not differ significantly between PVB and ICNB group within 4 h after surgery. This was in accordance with other studies that showed

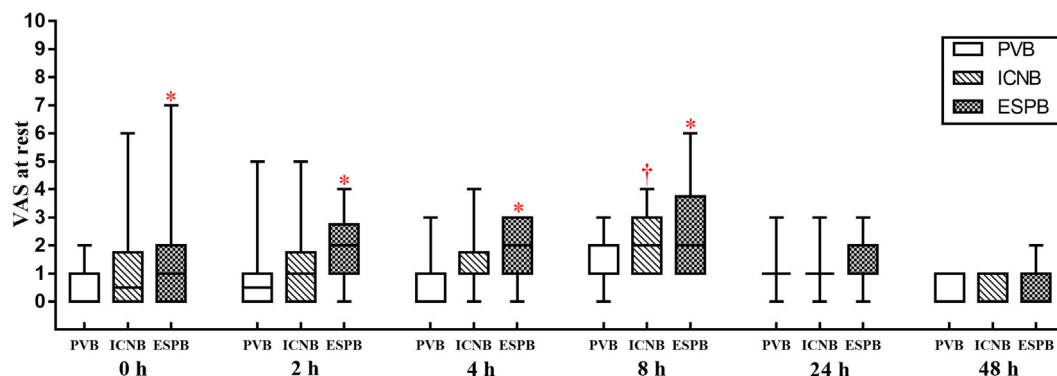


Fig. 2. Data are expressed as median (horizontal bar), interquartile range (box) and the maximum and minimum values (whiskers). All groups were compared using Kruskal-Wallis test and pairwise comparisons were analyzed using the Mann-Whitney *U* test. **p* < 0.0167 when ESPB compared with PVB. †*p* < 0.0167 when ICNB compared with PVB.

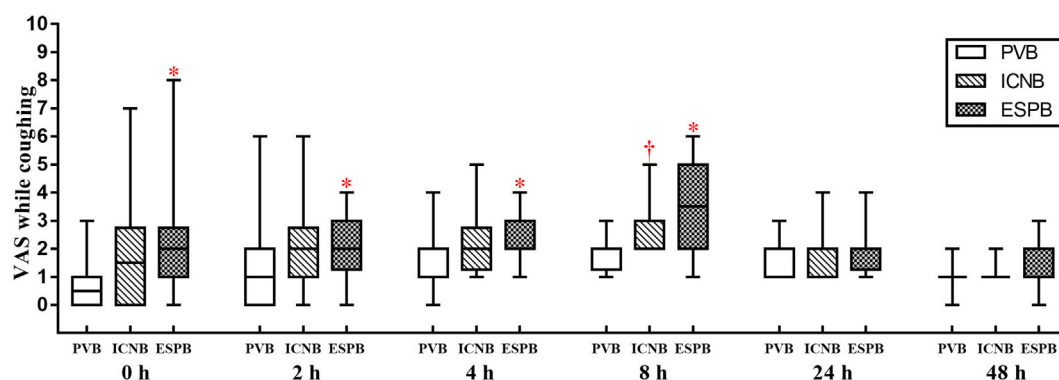


Fig. 3. Data are expressed as median (horizontal bar), interquartile range (box) and the maximum and minimum values (whiskers). All groups were compared using Kruskal-Wallis test and pairwise comparisons were analyzed using the Mann-Whitney U test. * $p < 0.0167$ when ESPB compared with PVB. † $p < 0.0167$ when ICNB compared with PVB.

Table 3

Rescue analgesia.

	PVB	ICNB	ESPB	p
Rescue analgesia requirement in PACU	0 (0%)	1 (4%)	5 (21%)	0.042
Rescue analgesia requirement, 1–8 h	2 (8%)	4 (17%)	5 (21%)	0.600
Rescue analgesia requirement, 0–24 h	3 (13%)	7 (29%)	11 (46%)	0.040
Rescue analgesia requirement, 0–48 h	3 (13%)	7 (29%)	11 (46%)	0.040

Values are presented as number (%).

ICNB was effective for providing analgesia during the early postoperative period [8,18–22]. In addition, Meierhenrich et al. [7] pointed out ICNB plus PCA could provide an excellent pain control at rest (pain scores < 4) for thoracotomy within four postoperative days, which was consistent with our study. However, in contrast with these studies, the lower VAS scores while coughing (< 4 at all observed time) in our study may be due to careful manipulation of surgeons, patient-controlled intravenous analgesia and different pain perception of patients in different culture backgrounds. A recent study [23] showed though paravertebral catheter infusions provided prolonged pain control and superior patient satisfaction compared with single-shot ICB after video-assisted thoracoscopic surgery, pain control during the first 24 h was similar. Similarly, our study also suggested ICNB associated with PCA is an effective postoperative analgesia regimen in thoracoscopic surgery, though its analgesic effect was slightly inferior to paravertebral block.

As reported by De Cassai et al. [24], a median volume of 3.4 ml of local anesthetic in ESPB could anesthetize one dermatome. At the meanwhile, referring to previous reports [13,25], we thought single-injection ESPB with 20 ml of 0.375% ropivacaine could provide extensive sensory blockade over hemithorax, which could completely cover pain area after thoracoscopic surgery. As a result, our study suggested single-injection ESPB provided inferior postoperative analgesia to multiple-injection PVB in thoracoscopic surgery as indicated by higher VAS scores, more morphine consumption and rescue analgesia, while single-injection ESPB and ICNB were equally effective in reducing pain. That our study failed to prove ESPB can be used as an effective alternative to PVB after thoracoscopic surgery could be explained by the possible acting mechanism of ESP block. Forero et al. [15] first demonstrated that local anesthetics in ESPB diffused anteriorly through the connective tissues and ligaments and entered paravertebral space where it acted on ventral and dorsal rami of the spinal nerve roots. Costache et al. [26] reported that the superior costovertebral ligament was not a barrier to the diffusion of the injectate and agreed to the hypothesis that the local anesthetics injected in ESPB could penetrate into the paravertebral space. Subsequently, most anatomical investigations [27,28] and clinical studies [29–31] supported the findings of Forero et al. However, a cadaveric study by Ivanusic et al. [32] suggested local anesthetics could not penetrate into the

paravertebral space and dorsal ramus were blocked due to posterior to the costovertebral foramen. They speculated potential involvement of lateral cutaneous branches of intercostal nerves lateral to the angle of the ribs because ESPB could provide analgesia for the hemithorax in many clinical reports. Taketa et al. [33] also reported that ESPB almost could not block the anterior branches of the intercostal nerves so that ESPB could provide insufficient but tolerable analgesia for patients undergoing thoracoscopic surgery. However, it was worth noting that lower volume and concentration of local anesthetic (15 ml of 0.2% levobupivacaine) in the study may also lead to this insufficient analgesia. Summarizing these studies, we thought local anesthetics in ESPB mainly produced extensive cranio-caudad spread along with erector spinae and were laterally distributed as an interfascial plane block [28] to involve the lateral cutaneous branch of intercostal nerves. Nevertheless, local anesthetics with higher volume and concentration could additionally diffused to the paravertebral space. However, the limited distribution of local anesthetics in the paravertebral space would cause the efficacy of ESPB to be weaker than that of PVB. As indicated in our study, when performing PVB and ESPB with the same volume and concentration of local anesthetic (20 ml of 0.375% ropivacaine), analgesic effect of ESPB was inferior to PVB. Besides, volume and concentration of local anesthetic in our study may be relatively lower for ESPB in thoracoscopic surgery because five patients in ESPB group expressed moderate pain and needed rescue analgesia immediately after surgery. However, in agreement with the results of most studies [9,13,14,34,35], our study implied that ESPB combined with a small amount of additional analgesic could reduce postoperative pain effectively within 48 postoperative hours after thoracoscopic surgery.

Previous studies showed a single-injection PVB could cover 3–6 dermatomal levels [36,37] and provided adequate analgesia for thoracic surgery. However, in this study, we decided to perform multiple-injection PVB based on the following considerations. First, single-injection PVB has a failure rate of 12%. In our study, successful paravertebral block was defined when pleural displacement was observed. But displacement of pleura is no guarantee for a successful paravertebral block. To decrease failure rate, we decided to perform triple-injection PVB. Second, Cheema et al. [37] reported that single-injection PVB produced unpredictable block. The aim of our study was to evaluate the analgesic effect of ESPB and ICNB compared with PVB and to determine whether ESPB was a good alternative to PVB on postoperative analgesia in thoracoscopic surgery. Therefore, paravertebral block should be sure to produce reliable coverage and provide good quality analgesia. Thirdly, although multiple-injection PVB is not recommended given increasing risks related to punctures, we believed serious complication could not occur in our study because multiple-injection PVB was performed under ultrasound guidance [38,39] and by an anesthesiologist with significant experience in these techniques. In our study, we found no significant difference in vessel injury among

the three groups and no other block-related complications took place. Certainly, we could not deny that our study was not adequately powered to detect these differences.

There are some limitations in this study. First, in order to minimize the patients' discomfort and make patients blinded to group allocations, all blocks were performed after general anesthesia. In this setting, displacement of pleura is no guarantee for a successful block. Second, recruitment of relatively healthy patients (ASA I and II) resulted in flawed extrapolation of the study. Thirdly, we did not use cutaneous sensory test to document the block range. Otherwise, we could make comparisons of the range of sensory block among ICNB, PVB and ESPB, which contributed to find the reasons for the difference of analgesic effect. Fourthly, time to block was not measured and the study was not adequately powered to detect block-related complications so that our study did not show the advantage of ESPB over PVB and ICNB in terms of safety and simplicity for postoperative analgesia in thoracoscopic surgery. In addition, it was possibly that 20 ml (75 mg) of ropivacaine was minimum volume (dose) for ESPB on postoperative analgesia in thoracoscopic surgery. Our results could provide a basis for future trials and the relationship between volume or concentration of LA with analgesic effect of ESPB should be further studied.

5. Conclusion

Ultrasound-guided ICNB and single-injection ESPB both provided inferior analgesia to multiple-injection PVB for thoracoscopic surgery, while no difference was found in postoperative analgesic effect between ICNB and single-injection ESPB. Besides, ICNB or single-injection ESPB in combination with PCA is also simple, safe and effective regimen for pain control after thoracoscopic surgery and can be used as a valid option especially when anesthesiologists have little experience in PVB or when PVB is contraindicated or failed. On the other hand, further studies are needed to determine the optimal volume and concentration of local anesthetics for ESPB.

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Declaration of Competing Interest

The authors declare no conflicts of interest.

References

- [1] Bendixen M, Jorgensen OD, Kronborg C, Andersen C, Licht PB. Postoperative pain and quality of life after lobectomy via video-assisted thoracoscopic surgery or anterolateral thoracotomy for early stage lung cancer: a randomised controlled trial. *Lancet Oncol* 2016;17:836–44.
- [2] Wildgaard K, Ravn J, Kehlet H. Chronic post-thoracotomy pain: a critical review of pathogenic mechanisms and strategies for prevention. *Eur J Cardiothorac Surg* 2009;36:170–80.
- [3] Yeung JH, Gates S, Naidu BV, Wilson MJ, Gao Smith F. Paravertebral block versus thoracic epidural for patients undergoing thoracotomy. *Cochrane Database Syst Rev* 2016;2:CD009121.
- [4] D'Ercole F, Arora H, Kumar PA. Paravertebral block for thoracic surgery. *J Cardiothorac Vasc Anesth* 2018;32:915–27.
- [5] Chelly JE. Paravertebral blocks. *Anesthesiol Clin* 2012;30:75–90.
- [6] Detterbeck FC. Efficacy of methods of intercostal nerve blockade for pain relief after thoracotomy. *Ann Thorac Surg* 2005;80:1550–9.
- [7] Meierhenrich R, Hock D, Kuhn S, Baltes E, Muehling B, Muche R, et al. Analgesia and pulmonary function after lung surgery: is a single intercostal nerve block plus patient-controlled intravenous morphine as effective as patient-controlled epidural analgesia? A randomized non-inferiority clinical trial. *Br J Anaesth* 2011;106:580–9.
- [8] Zhu M, Gu Y, Sun X, Liu X, Chen W, Miao C. Ultrasound-guided intercostal nerve block following esophagectomy for acute postoperative pain relief in the post-anesthesia care unit. *Pain Pract* 2018;18:879–83.
- [9] Adhikary SD, Pruett A, Forero M, Thiruvengataram V. Erector spinae plane block as an alternative to epidural analgesia for post-operative analgesia following video-assisted thoracoscopic surgery: a case study and a literature review on the spread of local anaesthetic in the erector spinae plane. *Indian J Anaesth* 2018;62:75–8.
- [10] Gurkan Y, Aksu C, Kus A, Yorukoglu UH, Kilic CT. Ultrasound guided erector spinae plane block reduces postoperative opioid consumption following breast surgery: a randomized controlled study. *J Clin Anesth* 2018;50:65–8.
- [11] Tulgar S, Kapakli MS, Senturk O, Selvi O, Serifsoy TE, Ozer Z. Evaluation of ultrasound-guided erector spinae plane block for postoperative analgesia in laparoscopic cholecystectomy: a prospective, randomized, controlled clinical trial. *J Clin Anesth* 2018;49:101–6.
- [12] Chin KJ, Adhikary S, Sarwani N, Forero M. The analgesic efficacy of pre-operative bilateral erector spinae plane (ESP) blocks in patients having ventral hernia repair. *Anaesthesia* 2017;72:452–60.
- [13] Forero M, Adhikary SD, Lopez H, Tsui C, Chin KJ. The erector spinae plane block: a novel analgesic technique in thoracic neuropathic pain. *Reg Anesth Pain Med* 2016;41:621–7.
- [14] Tsui BCH, Fonseca A, Munshey F, McFadyen G, Caruso TJ. The erector spinae plane (ESP) block: a pooled review of 242 cases. *J Clin Anesth* 2019;53:29–34.
- [15] Wang H, Li S, Liang N, Liu W, Liu H, Liu H. Postoperative pain experiences in Chinese adult patients after thoracotomy and video-assisted thoracic surgery. *J Clin Nurs* 2017;26:2744–54.
- [16] Vogt A, Stieger DS, Theurillat C, Curatolo M. Single-injection thoracic paravertebral block for postoperative pain treatment after thoracoscopic surgery. *Br J Anaesth* 2005;95:816–21.
- [17] Hill SE, Keller RA, Stafford-Smith M, Grichnik K, White WD, D'Amico TA, et al. Efficacy of single-dose, multilevel paravertebral nerve blockade for analgesia after thoracoscopic procedures. *Anesthesiology* 2006;104:1047–53.
- [18] Bolotin G, Lazarovici H, Uretzky G, Zlotnick AY, Tamir A, Sauter M. The efficacy of intraoperative internal intercostal nerve block during video-assisted thoracic surgery on postoperative pain. *Ann Thorac Surg* 2000;70:1872–5.
- [19] Taylor R, Massey S, Stuart-Smith K. Postoperative analgesia in video-assisted thoracoscopy: the role of intercostal blockade. *J Cardiothorac Vasc Anesth* 2004;18:317–21.
- [20] Concha M, Dagnino J, Cariaga M, Aguilera J, Aparicio R, Guerrero M. Analgesia after thoracotomy: epidural fentanyl/bupivacaine compared with intercostal nerve block plus intravenous morphine. *J Cardiothorac Vasc Anesth* 2004;18:322–6.
- [21] Perttunen K, Nilsson E, Heinonen J, Hirvisalo EL, Salo JA, Kalso E. Extradural, paravertebral and intercostal nerve blocks for post-thoracotomy pain. *Br J Anaesth* 1995;75:541–7.
- [22] Kadomatsu Y, Mori S, Ueno H, Uchiyama M, Wakai K. Comparison of the analgesic effects of modified continuous intercostal block and paravertebral block under surgeon's direct vision after video-assisted thoracic surgery: a randomized clinical trial. *Gen Thorac Cardiovasc Surg* 2018;66:425–31.
- [23] Hutchins J, Sanchez J, Andrade R, Podgaetz E, Wang Q, Sikka R. Ultrasound-guided paravertebral catheter versus intercostal blocks for postoperative pain control in video-assisted thoracoscopic surgery: a prospective randomized trial. *J Cardiothorac Vasc Anesth* 2017;31:458–63.
- [24] De Cassai A, Tonetti T. Local anesthetic spread during erector spinae plane block. *J Clin Anesth* 2018;48:60–1.
- [25] Hamilton DL, Manickam B. Erector spinae plane block for pain relief in rib fractures. *Br J Anaesth* 2017;118:474–5.
- [26] Costache I, de Neumann L, Ramnadan CJ, Goodwin SL, Pawa A, Abdallah FW, et al. The mid-point transverse process to pleura (MTP) block: a new end-point for thoracic paravertebral block. *Anaesthesia* 2017;72:1230–6.
- [27] Adhikary SD, Bernard S, Lopez H, Chin KJ. Erector spinae plane block versus retrolaminar block: a magnetic resonance imaging and anatomical study. *Reg Anesth Pain Med* 2018;43:756–62.
- [28] Onishi E, Toda N, Kameyama Y, Yamauchi M. Comparison of clinical efficacy and anatomical investigation between retrolaminar block and erector spinae plane block. *Biomed Res Int* 2019;2019:2578396.
- [29] Chin KJ, Malhas L, Perlas A. The erector spinae plane block provides visceral abdominal analgesia in bariatric surgery: a report of 3 cases. *Reg Anesth Pain Med* 2017;42:372–6.
- [30] Ueshima H, Hiroshi O. Spread of local anesthetic solution in the erector spinae plane block. *J Clin Anesth* 2018;45:23.
- [31] Schwartzmann A, Peng P, Maciel MA, Forero M. Mechanism of the erector spinae plane block: insights from a magnetic resonance imaging study. *Can J Anaesth* 2018;65:1165–6.
- [32] Ivanusic J, Konishi Y, Barrington MJ. A cadaveric study investigating the mechanism of action of erector spinae blockade. *Reg Anesth Pain Med* 2018;43:567–71.
- [33] Taketa Y, Irisawa Y, Fujitani T. Ultrasound-guided erector spinae plane block elicits sensory loss around the lateral, but not the parasternal, portion of the thorax. *J Clin Anesth* 2018;47:84–5.
- [34] Thiruvengataram V, Cruz Eng H, Adhikary SD. An update on regional analgesia for rib fractures. *Curr Opin Anaesthesiol* 2018;31:601–7.
- [35] Scimia P, Basso Ricci E, Droghetti A, Fusco P. The ultrasound-guided continuous erector spinae plane block for postoperative analgesia in video-assisted thoracoscopic lobectomy. *Reg Anesth Pain Med* 2017;42:537.
- [36] Naja ZM, El-Rajab M, Al-Tannir MA, Ziade FM, Tayara K, Younes F, et al. Thoracic paravertebral block: influence of the number of injections. *Reg Anesth Pain Med* 2006;31:196–201.
- [37] Cheema S, Richardson J, McGurgan P. Factors affecting the spread of bupivacaine in the adult thoracic paravertebral space. *Anaesthesia* 2003;58:684–7.
- [38] Baig S, Moon JY, Shankar H. Review of sympathetic blocks: anatomy, sonoanatomy, evidence, and techniques. *Reg Anesth Pain Med* 2017;42:377–91.
- [39] Krediet AC, Moayeri N, van Geffen GJ, Bruhn J, Renes S, Bigeleisen PE, et al. Different approaches to ultrasound-guided thoracic paravertebral block: an illustrated review. *Anesthesiology* 2015;123:459–74.